

Final URL: <https://sunkyus-eng.github.io/HW4-accessibility/>

AI Use: I do not have much experience with coding, so I used ChatGPT to help me understand accessibility concepts, troubleshoot HTML, CSS, and JavaScript, and work with VS Code and GitHub. I manually tested the final webpage and checked the accessibility changes myself.

Initial Automatic Accessibility Evaluation: I evaluated the original webpage using the WAVE Web Accessibility Evaluation Tool. WAVE detected 10 errors, 3 contrast errors, and 5 alerts. The main problems it identified included missing alternative text, missing form labels, low color contrast, no heading structure, and no page regions. WAVE was useful for finding several obvious accessibility problems, but some issues, such as keyboard interaction and the accessibility of the video content, required manual testing.

1. Headings and Landmarks

Problem:

The original webpage did not have a clear semantic heading or landmark structure. The “Links” and “Sign Up!” section titles looked like headings visually, but they were created using `<div>` elements instead of actual heading tags. The navigation and main content were also contained in generic `<div>` elements instead of semantic page regions.

Severity:

Medium

Why this is a problem:

Headings and landmarks help users understand the structure of a webpage. Screen-reader users can use headings and landmarks to move between important sections more quickly. Even if a page looks organized visually, assistive technology may not understand that structure unless semantic HTML is used.

What I changed:

I added an `<h1>` element for the main page title, “Accessibility Course Overview.” I changed the “Links” and “Sign Up!” section titles from `<div>` elements to `<h2>` elements. I also changed the navigation container to a `<nav>` element with `aria-label="Main navigation"` and wrapped the main webpage content inside a `<main>` element.

Detected by WAVE:

Yes. WAVE identified that the original webpage had no heading structure and no page regions.

Result after fix:

After making these changes, I ran WAVE again to check that the heading structure and page region issues were resolved. The page now has a clearer semantic structure that can be understood by both visual users and screen-reader users.

2. Good Alternative Text for Images

Problem:

The original webpage had several images that did not have appropriate alternative text. The three navigation images for Home, Links, and Sign Up did not have any alt text, so a screen-reader user would not know the purpose of those linked images. The decorative dot images were labeled "dot", even though they do not provide meaningful information. The play image for the video also did not have alternative text.

Severity:

High

Why this is a problem:

Images that communicate information or perform a function need a text alternative so users who cannot see the image can understand its purpose. Decorative images should not be announced because they add unnecessary information for screen-reader users. The assignment explains that visual information should have an alternative text form so that people who cannot see the content can still access the same information.

What I changed:

I added descriptive alternative text to the three navigation images: "Home", "Links", and "Sign Up". I changed the decorative dot images from alt="dot" to alt="" so screen readers will ignore them. The original custom play image was later removed when I replaced the original video player with the embedded YouTube video.

Detected by WAVE:

Partially. WAVE detected that the three linked navigation images were missing alternative text. However, automatic tools cannot always determine whether existing alternative text is actually useful. For example, the decorative dots already had alt="dot", so they technically had alternative text, but I manually determined that the dots were decorative and should instead use empty alternative text.

Result after fix:

After making these changes, I ran WAVE again to check that the missing alternative text errors for the navigation images were resolved.

3. Form Labels

Problem:

The original form had five controls without properly associated labels: First Name, Middle Initial, Last Name, Email, and Comments. Although some text appeared visually near the fields, it was not connected to the form controls using HTML `<label>` elements.

Severity:

High

Why this is a problem:

A screen reader needs a programmatic relationship between a form field and its label so the user knows what information should be entered. Visual placement alone does not provide that relationship. The assignment states that form labels are important for users who may not be viewing the page in its original visual layout.

What I changed:

I added a `<label>` for each of the five form controls and connected each label to its corresponding field using matching `for` and `id` attributes. I labeled the fields as First Name, Middle Initial, Last Name, Email, and Comments.

Detected by WAVE:

Yes. WAVE detected five missing form labels on the original webpage.

Result after fix:

After adding the labels, I ran WAVE again to check whether the five missing form label errors were resolved.

4. Video Captions

Problem:

The original video did not provide captions for the spoken audio.

Severity:

High

Why this is a problem:

Without captions, deaf or hard-of-hearing users may not be able to access information that is communicated through speech or other important sounds in the video.

What I changed:

I uploaded the video to YouTube and used YouTube's automatic captioning feature to create synchronized captions. I reviewed the captions for accuracy and embedded the YouTube video into the webpage so that users can turn captions on using the CC control.

Detected by WAVE:

No. WAVE could identify that video content was present, but I had to manually check whether captions were available and whether they accurately represented the audio.

Result after fix:

The spoken content of the video is now available as synchronized text, making the video more accessible to deaf and hard-of-hearing users.

5. Audio Descriptions

Problem:

The employment video contains important visual information in a graph that may not be fully communicated through the original audio. A blind or low-vision user could miss the employment percentages and comparison shown visually.

Severity:

High

Why this is a problem:

Important visual information in a video should also be available to users who cannot see the screen. Audio descriptions provide spoken descriptions of visual information that is not already communicated in the original audio.

What I changed:

I created an audio-described version of the video using YouDescribe. The audio description explains the important information shown in the employment graph, including the employment percentages for disabled and non-disabled Americans in 1991 and 2010. I added a link below the captioned video so users can access the audio-described version.

Detected by WAVE:

No. WAVE does not determine whether important visual information in a video has been adequately described, so I identified and tested this issue manually.

Result after fix:

Users who are blind or have low vision can now access the important visual information in the graph through the audio-described version of the video.

6. Tab Order & Visible Focus

Problem:

The original webpage removed the focus outline from the navigation links using `outline: none`. This made it difficult for keyboard users to see which interactive element was currently selected. I also needed to manually check whether the tab order followed a logical sequence through the webpage.

Severity:

High

Why this is a problem:

Users who navigate with a keyboard need to know which element currently has focus. If the focus indicator is hidden, a user may lose track of where they are on the page. The tab order should also follow the logical order of the content so that users can move through the page predictably.

What I changed:

I removed the CSS rule that hid the focus outline from navigation links. I added a visible black outline to links, inputs, text areas, elements with `tabindex`, and the embedded video whenever they receive keyboard focus. I then manually tested the page using the Tab and Shift+Tab keys to make sure the focus moved through the page in a logical order.

Detected by WAVE:

No. WAVE did not fully identify the visible-focus and tab-order problem. I tested this manually using the keyboard.

Result after fix:

Keyboard focus is now clearly visible as I move through interactive elements using Tab and Shift+Tab, and the page follows a logical navigation order.

7. Color Contrast

Problem:

Some links on the original webpage had insufficient color contrast. The links were displayed as gray text on a white background, which made them harder to distinguish and read. WAVE reported three contrast errors.

Severity:

Medium

Why this is a problem:

Low-contrast text can be difficult to read for users with low vision or reduced contrast sensitivity. Text and interactive elements should have enough contrast against their background to remain readable.

What I changed:

I changed the link color from gray to a darker blue that has stronger contrast against the white background. I also underlined the links so they are easier to identify as interactive elements.

Detected by WAVE:

Yes. WAVE detected three contrast errors on the original webpage.

Result after fix:

After changing the link color, I ran WAVE again to verify that the contrast errors were resolved.

8. Keyboard Access

Problem:

The original webpage contained interactive elements that were not fully keyboard accessible. The email preference control used an image as a custom checkbox instead of a native HTML checkbox. The original video also used a custom mouse-based play control.

Severity:

High

Why this is a problem:

Some users cannot use a mouse and navigate websites using a keyboard or another assistive input device that acts like a keyboard. If an interactive control only responds to mouse input, those users may not be able to use that function.

What I changed:

I replaced the image-based email preference control with a native HTML checkbox, which can be reached using Tab and activated using the Space key. The video had already been replaced with an embedded YouTube player that provides keyboard-accessible playback controls. I also used a native HTML button for the Sign Up control so it can be activated using the keyboard.

Detected by WAVE:

No. WAVE did not fully detect whether all webpage functionality could actually be operated with a keyboard. I tested this manually.

Result after fix:

I tested the webpage without using a mouse. I was able to navigate through the links and form fields using Tab, toggle the checkbox using Space, interact with the video controls, and activate the Sign Up button using the keyboard.

9. Information Conveyed Only Through Color

Problem:

The original form communicated validation errors mainly by changing the “Name” or “Email” text to red. A user who cannot easily distinguish colors might not notice that an error occurred or understand what needs to be fixed.

Severity:

Medium

Why this is a problem:

Color should not be the only way that important information is communicated. Users with color vision deficiencies or other visual impairments may not be able to recognize a change that is indicated only by color.

What I changed:

I added written error messages that appear when required information is missing. For example, the form now displays “Error: Please enter your first name, middle initial, and last name” or “Error: Please enter your email address.” I kept the visual color change as an additional cue, but color is no longer the only way the error is communicated. I also kept links underlined so they are identifiable by more than color alone.

Detected by WAVE:

Not completely. WAVE can identify some color-related accessibility issues, but determining whether information is being communicated only through color also requires manual evaluation.

Result after fix:

I tested the form by leaving required fields empty and activating the Sign Up button. A written error message now appears in addition to the visual color change, so the error can be understood without relying on color alone.

Overall Changes and Reflection

After completing all nine accessibility categories, I made changes across the HTML, CSS, JavaScript, and video content to make the webpage easier to use with different types of assistive technology.

The biggest changes were adding a clearer page structure with real headings and landmarks, adding useful alternative text to images, connecting form fields to proper labels, improving color contrast, and making keyboard focus easier to see. I also replaced some custom controls with standard HTML controls, such as using a real checkbox instead of an image-based checkbox, so the page works better without a mouse.

For the video, I added captions through YouTube and created an audio-described version using YouDescribe so that both the spoken and visual information are available in more than one format. I also changed the form validation so errors are not communicated only by turning text red. The form now shows written error messages and marks invalid fields more clearly.

Some problems were easy for WAVE to detect, such as missing alternative text, missing form labels, low color contrast, and missing heading structure. Other issues, such as keyboard access, visible focus, caption quality, audio descriptions, and information communicated only through color, needed to be tested manually.

Overall, the page still has the same basic purpose and layout, but the code now provides more information to assistive technology and gives users more than one way to understand and interact with the content.

Final WAVE Evaluation

After completing the accessibility changes, I ran WAVE again on the final webpage to compare it with the original evaluation.

Original results: 10 Errors / 3 Contrast Errors / 5 Alerts

Final results: 0 Errors / 0 Contrast Errors / 1 Alert: Youtube Video

The final evaluation showed that the automatically detectable accessibility problems were reduced. However, I still needed to manually test keyboard access, visible focus, captions, audio descriptions, and information communicated through color because WAVE cannot fully evaluate those areas.